Exam Date & Time: 22-Nov-2024 (02:30 PM - 05:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

7th Semester B.Tech - Biomedical Engineering
End Semester Examination - Nov 2024

Pattern Recognition [BME 4068]

Marks: 50 **Duration: 180 mins.**

Descriptive

Answer all the questions. Section Duration: 180 mins

Answer all questions.
Assume any missing data with proper reasoning.

1A) Based on the following dataset, using the Gaussian Naïve Bayes approach, determine the gender of a person having a height of 6 feet, weight of 130 lbs, and a foot size of 8 inches. (4)

Person	Height (feet)	Weight	Foot Size (Inches)
Male	6	180	12
Male	5.92	190	11
Male	5.58	170	12
Male	5.92	165	10
Female	5	100	6
Female	5.5	150	8
Female	5.42	130	7
Female	5.75	150	9

1B) A hospital wants to predict whether a patient will be readmitted within 30 days after discharge. They have data on factors such as age, comorbidities, length of stay, medication, and discharge instructions. Explain how logistic regression could be used in (3)

this context to predict readmission, including how it might handle binary outcomes, interpret coefficients, and help in making medical decisions.

1C) A square matrix is skew-symmetric if $A^T = -A$. If A is an n*n skew-symmetric matrix, with n being odd, establish that det (A) = 0. (3)

2A) Given the dataset {A, B, C, D, E} and the following distance matrix, construct a dendrogram using the single linkage hierarchical clustering using the agglomerative method. (4)

Sl.No.	A	B	C	D	E
A	0	9	3	6	11
B	9	0	7	5	10
C	3	7	0	9	2
D	6	5	9	0	8
E	11	10	2	8	0

2B) Calculate the weights required to perform the following classification using the perceptron network. The vectors (1, 1, 1, 1) and (-1, 1, -1, -1) belong to the class (so have a target value 1), and vectors (1, 1, 1, -1) and (1, -1, -1, 1) are not belonging to the class (so have a target value -1). Assume the learning rate as 1 and the initial weights as 0. (4)

2C) You are given a large dataset of gene expression levels across thousands of genes for different types of cancer cells, and you need to identify patterns to distinguish between cancer types. Analyze how you would apply PCA to reduce the dimensionality of this dataset. Evaluate methods to validate that the reduced dimensions retain sufficient information to distinguish between cancer types effectively. (2)

3A) Explain the different steps involved in training a typical radial basis function (RBF) neural network algorithm. (4)

3B) A consulting firm rents cars from three rental agencies in the following manner: 20% from agency D, 20% from agency E, and 60% from agency F. If 10% of cars from D, 12% cars from E, and 4% cars from F have bad tyres, what is the probability that the firm will get a car with bad tyres? Determine the probability that a car with bad tyres is rented from agency F. (3)

3C) Imagine you are a biomedical engineer developing a diagnostic model for a hospital to predict malfunctions in critical medical devices, such as ventilators or dialysis machines. The model uses a decision tree to classify devices as "low risk" or "high risk" for malfunction based on factors like operational time, recent error codes, environmental conditions (e.g., temperature and humidity), and maintenance history. (3)

- Analyse the benefits and drawbacks of using a decision tree in this context, especially in terms of interpretability, real-time decision-making, and the model's ability to handle complex relationships among factors.
- Critique how decision trees might fall short in detecting rare but critical malfunction patterns and discuss the impact this limitation could have on patient safety and device reliability.

4A) Consider an 8-character password made up of only uppercase letters (26 options) and digits (10 options). Determine the probability that the password contains two letters that are the same exactly and three digits that are the same exactly, with no character repeating. (4)

4B) Consider the five data points: P1 (1,2,3); P2 (0,1,2); P3 (3,0,5); P4 (4,1,3); P5 (5,0,1). Apply the K-means clustering algorithm to group these data points into two clusters using the Manhattan distance metric. (4)

4C) Briefly explain the purpose of the learning rate in the gradient descent algorithm and how it affects the convergence of the model. (2)

5A) Apply an Adaline network to train the ANDNOT function with bipolar inputs and targets. Perform two epochs of training. The truth table for the ANDNOT function with bipolar inputs and targets is as given below: (Assume the initial weights, bias, and learning rate as 0.2) (4)

X_1	X_2	1	t
1	1	1	-1
1	-1	1	1
-1	1	1	-1
-1	-1	1	-1

5B) Given the dataset shown below, apply the methods of ordinary least squares/linear regression to find the equation for the prediction of the student’s end-term exam grade (3)

based on the student's mid-term exam grade in the course. Also, predict the final exam grade of a student who scored 86 in the mid-term exam.

'X' mid-term	'Y' end-term
72	84
50	63
81	77
74	78
94	90
86	75
59	49
83	79
65	77
33	52
88	74
81	90

5C) You are a biomedical engineer working on a project to develop an artificial neural network (ANN) to predict the risk of a specific disease (e.g., cardiovascular disease) based on a set of input biomarkers, such as cholesterol levels, blood pressure, and genetic factors. The model uses backpropagation to optimize the weights during training. (3)

- Evaluate how bias in the input data (e.g., demographic imbalances or measurement errors in biomarkers) could propagate through the ANN during training via backpropagation. What strategies would you recommend to detect and mitigate bias in the learned model, ensuring fairness across diverse patient groups?

- Assess how backpropagation affects the model's ability to learn from underrepresented classes and how class weighting could influence the network's convergence and final performance.

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